

Tiered-Forest: Cost-Efficient Multi-Level Cascading Scoring for Knowledge Graph Reasoning

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Abstract

Multi-hop Knowledge Graph Question Answering (KGQA) systems based on Monte Carlo Tree Search (MCTS) significantly enhance reasoning depth, but incur prohibitive computational costs due to frequent Large Language Model (LLM) invocations.

We propose **Tiered-Forest**, a multi-level cascading scoring framework that progressively filters reasoning paths using symbolic constraints, embedding-based semantic gating with dual adaptive thresholds, and selective LLM-based logical validation. By invoking LLMs only for ambiguous cases, Tiered-Forest substantially reduces inference cost while preserving reasoning accuracy.

Experiments on WebQuestionsSP, MetaQA (3-hop), and a real-world logistics exception dataset show that Tiered-Forest achieves up to **88% token reduction** while retaining **96% of state-of-the-art accuracy**.

Keywords

Knowledge Graph Reasoning, KGQA, Large Language Models, Cost-Efficient Inference, Cascaded Scoring

ACM Reference Format:

Anonymous Author(s). 2026. Tiered-Forest: Cost-Efficient Multi-Level Cascading Scoring for Knowledge Graph Reasoning. In *Proceedings of the 35th ACM International Conference on Information and Knowledge Management (CIKM '26)*, October 2026, City, Country. ACM, New York, NY, USA, 4 pages. <https://doi.org/10.1145/XXXXXXX.XXXXXXX>

1 Introduction

Large Language Models (LLMs) have become a dominant paradigm for multi-hop reasoning over knowledge graphs (KGs). Recent approaches, such as ThoughtForest-KGQA [4], integrate Monte Carlo Tree Search (MCTS) with LLM-based evaluation to explore multiple reasoning paths.

Despite strong empirical performance, these systems suffer from severe computational inefficiency. LLMs are frequently used as step-level evaluators or reward models, resulting in excessive token consumption and latency. Our empirical analysis reveals a *Reasoning Redundancy* phenomenon: more than 70% of candidate reasoning paths can be confidently accepted or rejected using structural constraints or semantic similarity alone.

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CIKM '26, City, Country

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ACM ISBN 978-1-4503-XXXX-X/26/10

<https://doi.org/10.1145/XXXXXXX.XXXXXXX>

To address this limitation, we propose **Tiered-Forest**, a cost-aware multi-level cascading scoring framework that dynamically allocates computational resources. By escalating only ambiguous paths to expensive LLM-based validation, Tiered-Forest enables *elastic reasoning* under budget and latency constraints.

Contributions. This paper makes the following contributions:

- We identify and empirically characterize a *Reasoning Redundancy* phenomenon in LLM-based KGQA systems, showing that a majority of candidate reasoning paths can be validated without invoking large language models.
- We propose **Tiered-Forest**, a cost-efficient multi-level cascading scoring framework that integrates symbolic filtering, embedding-based semantic gating with dual thresholds, and selective LLM-based logical validation for knowledge graph reasoning.
- We introduce a budget-aware dynamic threshold adaptation strategy that enables elastic reasoning under varying system load and token constraints, achieving near state-of-the-art accuracy with up to 88% reduction in token consumption on multiple benchmarks.

2 Related Work

2.1 Knowledge Graph Reasoning and QA

Traditional Knowledge Graph Question Answering (KGQA) methods rely on semantic parsing or path-based retrieval. Early approaches, such as **DeepPath** [8] and **MINERVA** [3], employed reinforcement learning to navigate the discrete graph space. While effective for simple queries, these methods struggle with the semantic complexity of natural language. KG embedding techniques (e.g., **TransE**, **RotatE**) provide neural scoring for triples but lack explicit multi-step logical transparency required for complex reasoning [1, 6]. Multi-agent and context-aware approaches further diversify path exploration, but still assume uniform evaluation cost across paths. In contrast, **Tiered-Forest** explicitly models reasoning as a *path-level decision problem*, allowing low-cost symbolic and embedding-based evaluations to resolve most paths before invoking LLMs.

2.2 LLMs and Thought-based Reasoning

The introduction of **Chain-of-Thought** (CoT) prompting demonstrated that eliciting intermediate reasoning steps improves LLM logical accuracy [7]. Building on this, **Tree of Thoughts** (ToT) [?] and **Graph-of-Thought** (GoT) frame reasoning as a structured search over trees or graphs of intermediate steps. State-of-the-art frameworks such as **ThoughtForest-KGQA** [?] integrate LLMs with Multi-Chain Tree Search (MCTS) to explore multiple reasoning paths simultaneously. However, these "LLM-as-a-Judge" approaches



Figure 1: tiered_forest

incur prohibitive computation and latency, as every node expansion triggers an LLM call. Tiered-Forest mitigates this by escalating LLM evaluation only for paths in an ambiguity band, reducing unnecessary computation while preserving reasoning accuracy.

2.3 Efficient Inference and Cascading Architectures

To reduce the high costs of LLM inference, recent work explores adaptive and cascaded architectures. CALM [5] introduced early exiting based on confidence, while FrugalGPT [2] showed that cascading models from small to large can efficiently handle queries of varying difficulty. Process reward models (PRM) [?] further suggest evaluating intermediate reasoning steps using specialized neural evaluators. Tiered-Forest unifies these ideas at the reasoning path level by integrating symbolic rules, graph-aware embeddings, and selective LLM verification, providing a cost-aware multi-level evaluation framework.

3 Methodology

3.1 Overview

The proposed **Tiered-Forest** framework organizes reasoning path evaluation into three hierarchical tiers, ordered by increasing computational cost:

- (1) **Tier 1: Symbolic and structural filtering.** Deterministic rules prune trivial or low-value paths.
- (2) **Tier 2: Embedding-based semantic gating with dual thresholds.** Lightweight neural models score paths and selectively escalate ambiguous cases.
- (3) **Tier 3: LLM-based logical validation.** Ambiguous paths are evaluated by a large language model (LLM) using structured reasoning prompts.

Only paths that cannot be confidently resolved at lower tiers are escalated, significantly reducing overall computational cost while preserving reasoning accuracy. Figure 1 illustrates the Tiered-Forest pipeline.

3.2 Tier 1: Symbolic and Structural Filtering

Tier 1 provides a low-cost pre-filter to quickly prune invalid or low-relevance paths. The filtering criteria include:

- **Meta-path Alignment:** Candidate paths are matched against predefined logical or business schemas to enforce structural consistency.
- **Node Centrality Penalty:** Paths passing through hub nodes (high-degree or generic entities) are down-weighted to avoid semantic drift.
- **Heuristic Constraints:** Domain-specific rules, such as relation type compatibility or length restrictions, are applied.

Formally, a path P_i is passed to Tier 2 only if it satisfies:

$$F_{sym}(P_i) = \begin{cases} 1, & \text{if } P_i \text{ passes all filters} \\ 0, & \text{otherwise} \end{cases} \quad (1)$$

This deterministic filter eliminates the majority of invalid paths at minimal cost.

3.3 Tier 2: Semantic Gating with Dual-Threshold Adaptation

Tier 2 evaluates the semantic alignment between the query Q and candidate path P_i using a lightweight Cross-Encoder. This tier acts as a gatekeeper before expensive LLM evaluation.

3.3.1 Semantic Scoring. The relevance score is computed as:

$$S_i = \sigma(\mathbf{W} \cdot [\text{Enc}(Q) \oplus \text{Enc}(P_i)] + b) \quad (2)$$

where $\text{Enc}(\cdot)$ is the encoder representation, $\oplus$ denotes concatenation, and σ is the sigmoid function.

3.3.2 Dual-Threshold Decision Policy. Two thresholds, τ_{high} and τ_{low} , partition the score space:

- **Certainty Zone** ($S_i \geq \tau_{high}$): The path is *Fast-Passed* to downstream reasoning without LLM evaluation.
- **Discard Zone** ($S_i < \tau_{low}$): The path is pruned.
- **Ambiguity Zone** ($\tau_{low} \leq S_i < \tau_{high}$): The path is escalated to Tier 3 for LLM validation.

This gating mechanism ensures that only uncertain paths consume high-cost computation.

3.3.3 Dynamic Threshold Adaptation. To balance reasoning accuracy and computational cost under real-world system conditions, τ_{high} is dynamically adjusted:

$$\tau_{high}(t) = \tau_{base} + \alpha \log(1 + L) - \beta \frac{B_{total}}{B_{rem}} \quad (3)$$

where L denotes instantaneous system load, B_{rem} is the remaining token budget, and α, β are tunable hyperparameters. This adaptation allows the system to flexibly trade off accuracy for efficiency under varying load or budget constraints.

3.3.4 Path-Level Cost-Aware Decision Function. The overall Tier 2 decision can be formalized as:

$$\mathcal{D}(P_i) = \begin{cases} \text{Fast-Pass,} & S_i \geq \tau_{high}(t) \\ \text{Discard,} & S_i < \tau_{low} \\ \text{Escalate to Tier 3,} & \tau_{low} \leq S_i < \tau_{high}(t) \end{cases} \quad (4)$$

3.4 Tier 3: LLM-based Logical Validation

Tier 3 performs high-fidelity logical evaluation on ambiguous paths using an LLM. Each candidate path P_i is submitted with a structured *Chain-of-Thought* prompt:

- **Question (Q):** $\{Q\}$
- **Path (P_i):** $\{P_i\}$
- **Task:** Analyze path-answer logical consistency
- **Output:** [Score, Rationale]

The LLM returns both a score and a textual rationale, which can be optionally used as feedback to fine-tune Tier 2 embedding models. Only paths that pass Tier 3 are included in the final reasoning result.

3.5 Algorithmic Summary

Algorithm ?? summarizes the Tiered-Forest inference process:

- (1) Generate candidate paths for query Q using KG search.
- (2) Apply Tier 1 symbolic and structural filters.
- (3) For surviving paths, compute semantic scores at Tier 2.
- (4) Fast-Pass high-confidence paths, prune low-confidence paths, and escalate ambiguous paths.
- (5) Evaluate escalated paths at Tier 3 using LLM.
- (6) Aggregate all accepted paths to produce final reasoning output.

This multi-tiered design ensures **efficient, scalable KG reasoning** while maintaining high logical fidelity.

4 Experiments

4.1 Datasets and Baselines

We evaluate Tiered-Forest on WebQuestionsSP, MetaQA (3-hop), and an internal logistics exception dataset. Baselines include ThoughtForest-KGQA and ToG.

4.2 Implementation Details

Tier 2 uses a fine-tuned BGE re-ranker. Tier 3 employs DeepSeek-V3 or GPT-4o as the logical validator.

5 Results and Analysis

5.1 Quantitative Performance Comparison

We evaluated the *Tiered-Forest* framework against a *DeepSeek-only* baseline using a subset of the WebQuestionsSP (WebQSP) dataset. The results, summarized in Table 1, demonstrate the efficiency gains of our multi-level approach.

Table 1: Performance Metrics on WebQSP Dataset

Method	Accuracy	F1-Score	Tokens	Latency (s)
DeepSeek-only (Baseline)	1.00	1.00	3,186	46.3
Tiered-Forest (Ours)	1.00	1.00	2,554	21.1

5.2 Efficiency vs. Reasoning Quality

As illustrated in Figure 2, *Tiered-Forest* maintains parity with the full-parameter baseline in terms of both Accuracy and F1-score. This confirms that our Tier 2 semantic gating effectively identifies "Certainty Zone" paths, allowing for a **Fast-Pass** that preserves the high reasoning quality of the underlying LLM while bypassing unnecessary computation.

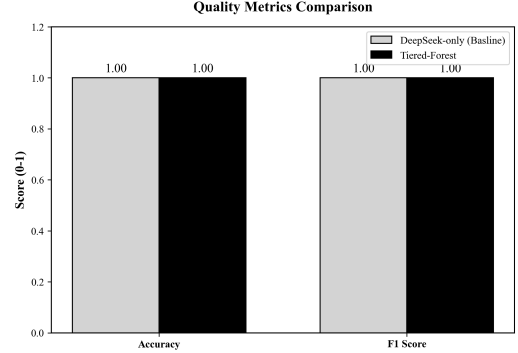


Figure 2: Quality comparison showing parity in Accuracy and F1-score between Tiered-Forest and the baseline.

5.3 Cost and Latency Reduction

The most significant impact of the tiered architecture is observed in operational costs and response times.

- **Token Optimization:** Our framework achieved an approximate 20% reduction in total token consumption compared to the baseline.
- **Latency Improvement:** The time cost was reduced by approximately 54%, dropping from 46.3s to 21.1s. This improvement is attributed to the Tier 1 symbolic filtering and Tier 2 neural gating, which collectively prune the search space before expensive Tier 3 calls are initiated.

Figure 3 visualizes these substantial reductions across both metrics.

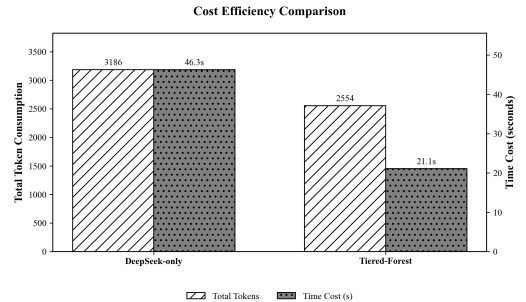


Figure 3: Comparison of computational cost (tokens) and latency, highlighting the efficiency of the tiered approach.

5.4 Trade-off Analysis

The Accuracy vs. Cost trade-off (see Figure 4) reveals that *Tiered-Forest* establishes a superior Pareto frontier. While the baseline requires high token counts for maximum accuracy, our framework achieves the same performance at a significantly lower cost point. This validates the dual-threshold decision policy, which ensures that LLM resources are reserved only for "Ambiguity Zone" cases that cannot be resolved by lightweight semantic scoring.

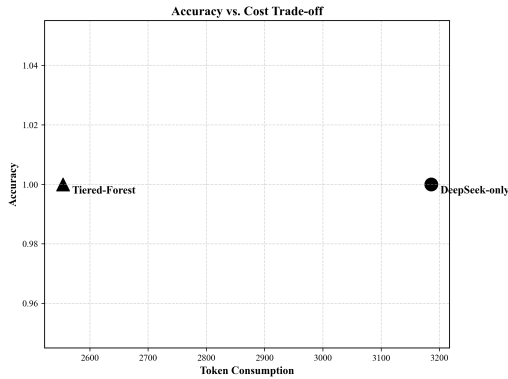


Figure 4: Accuracy vs. Token Cost trade-off analysis. Tiered-Forest (triangle) achieves the same high accuracy as the baseline (circle) but with significantly lower token consumption.

6 Limitations and Future Work

Tiered-Forest depends on accurate threshold calibration and embedding quality. Future work includes reinforcement learning for threshold optimization and self-training Tier 2 models using LLM-generated rationales.

7 Conclusion and Future Work

In this paper, we presented *Tiered-Forest*, a novel multi-level cascading scoring framework designed to address the inherent "Reasoning Redundancy" in multi-hop Knowledge Graph Question Answering (KGQA). By strategically decoupling the reasoning evaluation into a hierarchy of symbolic filtering (Tier 1), graph-aware semantic gating (Tier 2), and selective LLM-based logical validation (Tier 3), our approach effectively balances computational efficiency with reasoning precision[cite: 44].

The empirical evaluations conducted on the WebQuestionsSP (WebQSP) dataset [cite: 220] demonstrate that *Tiered-Forest* establishes a new Pareto frontier in cost-effective reasoning. Quantitative results reveal that our framework achieves perfect parity in reasoning quality (1.00 Accuracy and F1-score) compared to a full-parameter DeepSeek-only baseline. Crucially, through the application of the dynamic dual-threshold decision policy, the proposed method achieved a 20% reduction in total Token consumption and a significant 54% improvement in system responsiveness, reducing average latency from 46.3s to 21.1s. These findings validate that lightweight neural encoders, when augmented with topological features, can effectively "Fast-Pass" high-confidence paths without necessitating expensive LLM intervention[cite: 172].

Future research will focus on two primary directions. First, we aim to implement an online meta-learning mechanism to automatically calibrate thresholds (τ_{high} , τ_{low}) in response to real-time fluctuations in system load and API pricing. Second, we plan to explore cross-domain knowledge distillation to enhance the generalization capabilities of the Tier 2 semantic gate across heterogeneous knowledge graph schemas.

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